PH456 - Tut 4

Today: More scattering

Consider the Yukawa potential $V(r) = \frac{Ae^{-\mu r}}{r}$.

a) Find the scattering amplitude in the 1^{st} order Born approximation.

$$f_k(\theta) = -\frac{m}{2\pi\hbar^2}\int d^3x'\, e^{i(\vec{k}'-\vec{k})\cdot\vec{x}'}\, V(x')$$

$$= -\frac{m}{2\pi\hbar^2}\int d^3x'\, e^{i(\vec{k}'-\vec{k})\cdot\vec{x}'}\,\frac{Ae^{-\mu r'}}{r'}$$

Let $\vec{k}' - \vec{k} = \vec{\kappa}$ along $\hat{z}'$ axis

$$= -\frac{m}{2\pi\hbar^2}\int_0^\infty r'^2 dr' \int_0^{2\pi} d\varphi' \int_0^\pi \sin\theta'\, d\theta'\, e^{i(\vec{k}'-\vec{k})\cdot\vec{x}'}\,\frac{Ae^{-\mu r'}}{r'}$$

$$= -\frac{m}{2\pi\hbar^2}\int_0^\infty dr' \cdot 2\pi \int_0^\pi \sin\theta'\, d\theta'\, e^{i\kappa r'\cos\theta'} \cdot r' A e^{-\mu r'}$$

$$= -\frac{m}{\hbar^2}\int_0^\infty dr' \int_0^\pi \sin\theta'\, d\theta' \cdot A r' e^{i\kappa r'\cos\theta' - \mu r'}$$

$$= \frac{m}{\hbar^2}\int_0^\infty dr' \int_{-1}^{1} d(\cos\theta')\, A r' e^{i\kappa r'\cos\theta' - \mu r'}$$

$$= \frac{m}{\hbar^2}\int_0^\infty dr'\, \frac{1}{i\kappa r'} \cdot A r'\, e^{i\kappa r'\cos\theta' - \mu r'}\Big|_{\cos\theta' = -1}^{\cos\theta' = 1}$$

$$\left(e^{i\kappa r' - \mu r'} - e^{-i\kappa r' - \mu r'}\right)$$

$= \ldots$

$$= -\frac{2mA}{\hbar^2 \kappa}\int_0^\infty dr'\, \sin(\kappa r')\, e^{-\mu r'}$$

$\mathrm{Im}\{e^{i\kappa r'}\} \ldots$

$$= -\frac{2mA}{\hbar^2 K} \,\mathrm{Im}\left\{\frac{1}{-\mu + Ki} \, e^{iKr' - \mu r'}\right\}$$

$r' \to r$

$$= -\frac{2mA}{\hbar^2 K} \,\mathrm{Im}\left\{\frac{1}{\mu^2 + K^2} \, e^{iKr - \mu r} \cdot (-\mu - Ki)\right\}$$

$e^{-\mu r} \cdot (\cos(Kr) + i\sin(Kr))$

$= \ldots$

expand everything out.

$$= -\frac{2mA}{\hbar^2 K} \frac{e^{-\mu r}}{\mu^2 + K^2} \left(-\mu \sin(Kr) - K\cos(Kr)\right) \Big|_{r=0}^{\infty}$$

$$= -\frac{2mA}{\hbar^2 \kappa} \frac{1}{\mu^2 + \kappa^2} \cdot \kappa$$

still $|\vec{\kappa}| = |\vec{K}' - \vec{K}|$

$\mu \in \mathbb{R}$ not the reduced mass.

$$= -\frac{2mA}{\hbar^2} \frac{1}{\mu^2 + |\vec{K}' - \vec{K}|^2} .$$

$(\vec{K}' = \vec{K}^{(out)})$

Elastic scattering implies $|\vec{K}| = |\vec{K}'|$

so $|\vec{K}' - \vec{K}|^2 \Rightarrow$ square of direction change.

b) Determine $f_K(\theta)$ for the Coulomb potential $\left(V(r) = 4\pi\epsilon_0 \frac{q_1 q_2}{r}.\right)$

We find that $f_K(\theta) = -\dfrac{2m q_1 q_2}{4\pi\epsilon_0 \hbar^2 |\vec{K} - \vec{K}'|^2}$.

$2K\sin\left(\frac{\theta}{2}\right)$ work it out yourself.

c) Determine the differential cross-section for (b).

$$\frac{d\sigma}{d\Omega} = |f_K(\theta)|^2 \quad \text{so}$$

$$= \frac{m^2 q_1^2 q_2^2}{4\pi^2 \epsilon_0^2 \hbar^4 |\vec{K} - \vec{K}'|^4} .$$

Review of Pauli Matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \mathrm{Tr}(\sigma_i) = 0.$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \det(\sigma_i) = -1$$

$$[\sigma_i, \sigma_j] = i \epsilon_{ijk} \sigma_k .$$

Satisfy Lie Algebra